7. LLM Preference Alignment and Reasoning

CSC3043S Reasoning in AI: Large Language Models

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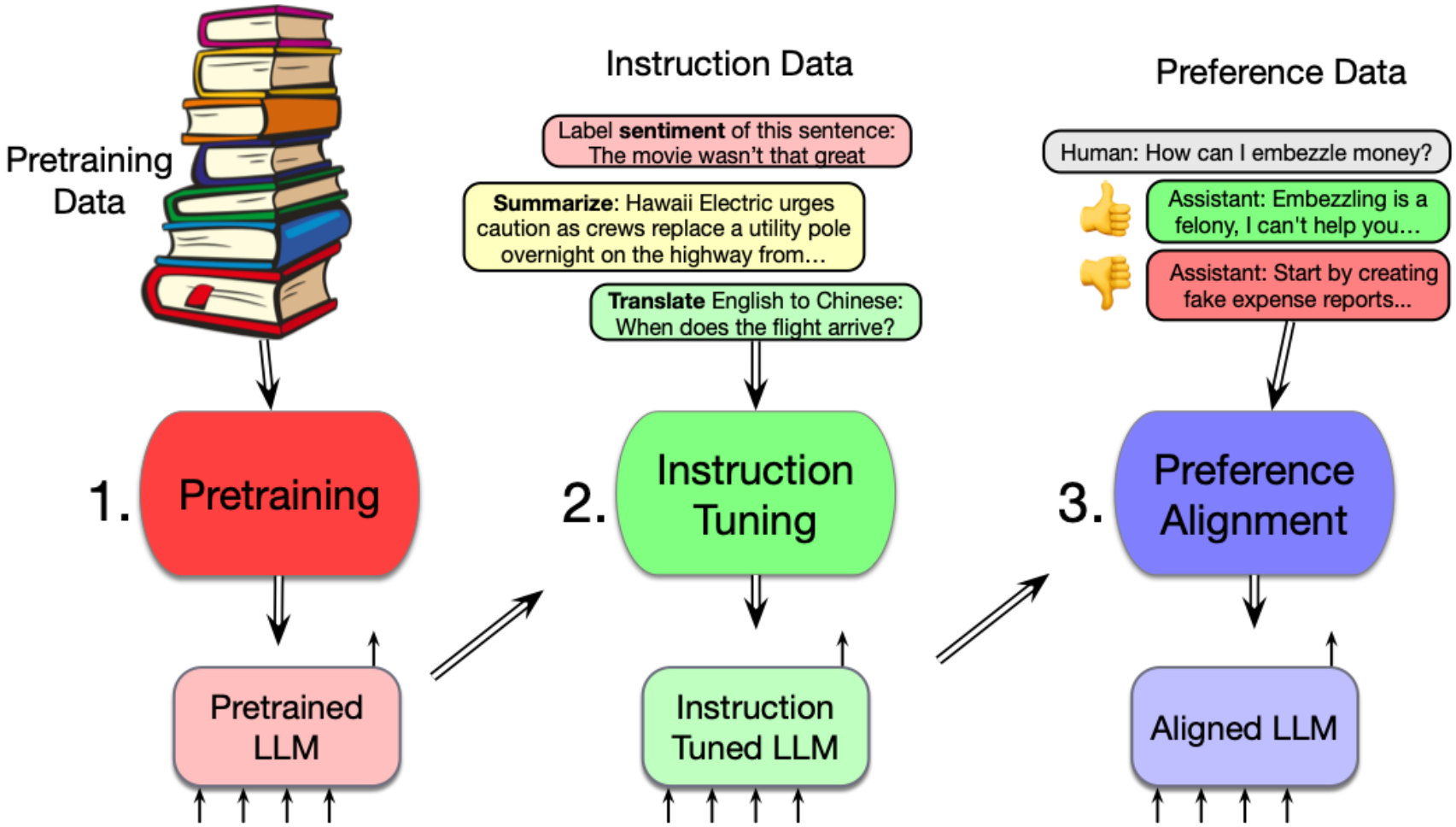
Slides from Jurafsky and Martin, Tatsunori Hashimoto (Stanford), others used as cited.



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Three stages of LLM training



Language model alignment

- Language models are not aligned with human values

PROMPT	*It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*
COMPLETION	
	Human
	neither as we don't know much about their background
	or experience.

- An AI is "aligned" if it is **helpful**, **honest** and **harmless** (Askell et al., 2020)

[Ethical-Advice Taker: Do Language Models Understand Natural Language Interventions?, Zhao et al. 2021]

Language model alignment

- People don’t always write (as annotators) what they prefer as LM output
  - Easier to collect ratings of outputs or preferences between possible outputs
- Supervised instruction fine-tuning does not control the length and style of outputs well enough

Optimizing for human preferences

- For every output sample *s* of a LM on some task, obtain a human reward R(*s*) – higher is better
- Example: Summarization

SAN FRANCISCO, California (CNN) -- A magnitude 4.2 earthquake shook the San Francisco ... overturn unstable objects.

An earthquake hit San Francisco. There was minor property damage, but no injuries.

s_1

$R(s_1) = 8.0$

The Bay Area has good weather but is prone to earthquakes and wildfires.

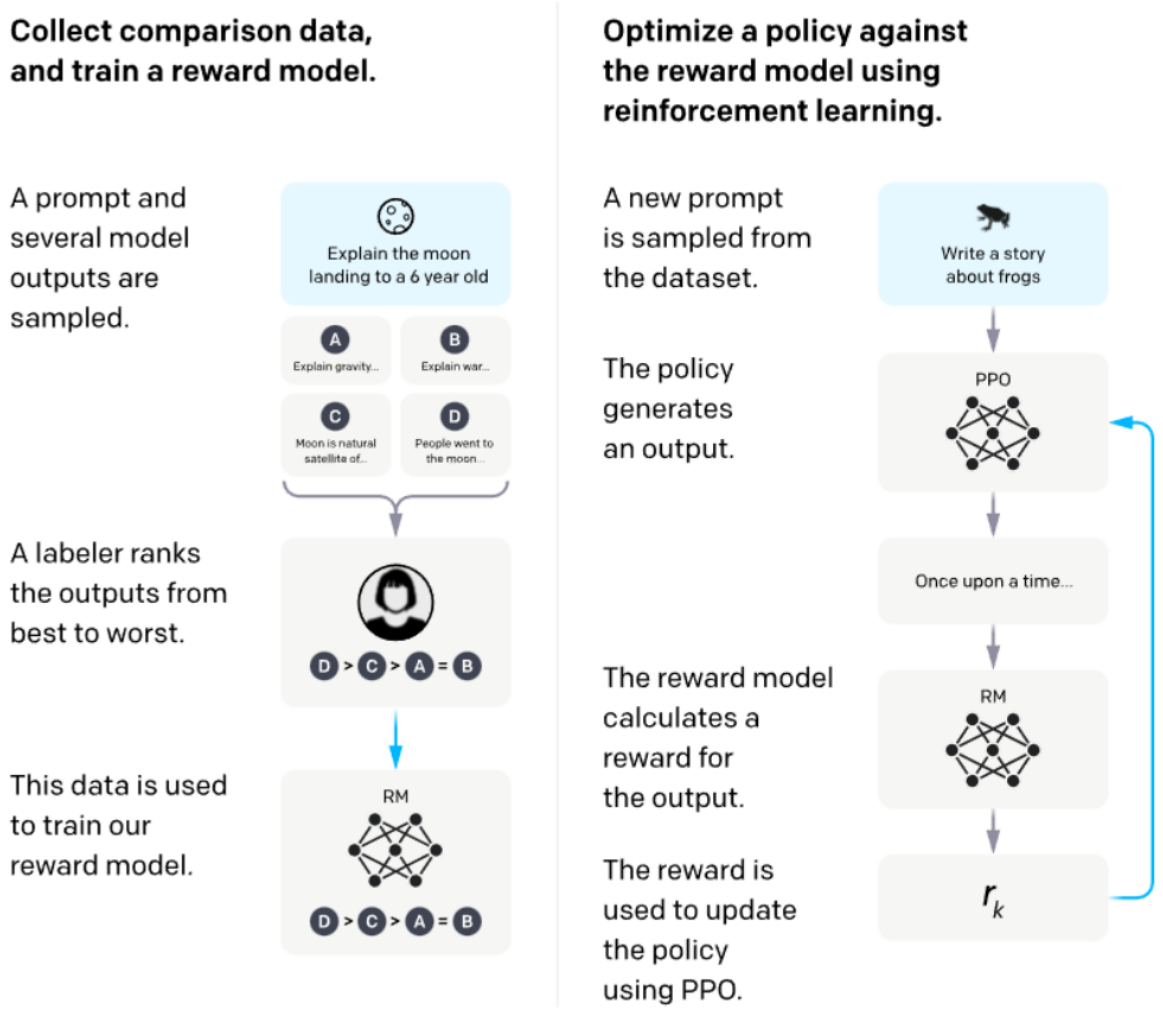
s_2

$R(s_2) = 1.2$

Reinforcement Learning from Human Feedback (RLHF)

1. Collect instruction / response data and train a supervised model (*policy*)
   - Supervised instruction fine-tuning
2. Collect human preference data ranking or scoring multiple possible responses to instructions
   - This data is used to train a r**eward model** to emulate the human judgements
3. Optimize a policy against the reward model using **reinforcement learning**
   - Generate responses using the current policy (the LLM)
   - Score responses with the reward model
   - Use reinforcement learning to update the policy

Reinforcement Learning from Human Feedback (RLHF)



Ouyang 2022

Reinforcement Learning from Human Feedback (RLHF): Reward modelling

- Human-in-the-loop is expensive: rather model human preferences as a separate problem
- Train a reward model LM to predict human preferences from an annotated dataset, then use RL to optimize the policy (main LM) for the reward model instead
- Pairwise comparisons can be more reliable than asking for direct ratings

An earthquake hit San Francisco. There was minor property damage, but no injuries. > A 4.2 magnitude earthquake hit San Francisco, resulting in massive damage. > The Bay Area has good weather but is prone to earthquakes and wildfires.

https://huggingface.co/blog/rlhf

Reinforcement Learning from Human Feedback (RLHF): Reward modelling

- Loss for pairwise comparison model

$$J_{RM}(\phi) = -\mathbb{E}_{(s^w, s^l)\sim D}\left[\log \sigma(RM_\phi(s^w) - RM_\phi(s^l))\right]$$

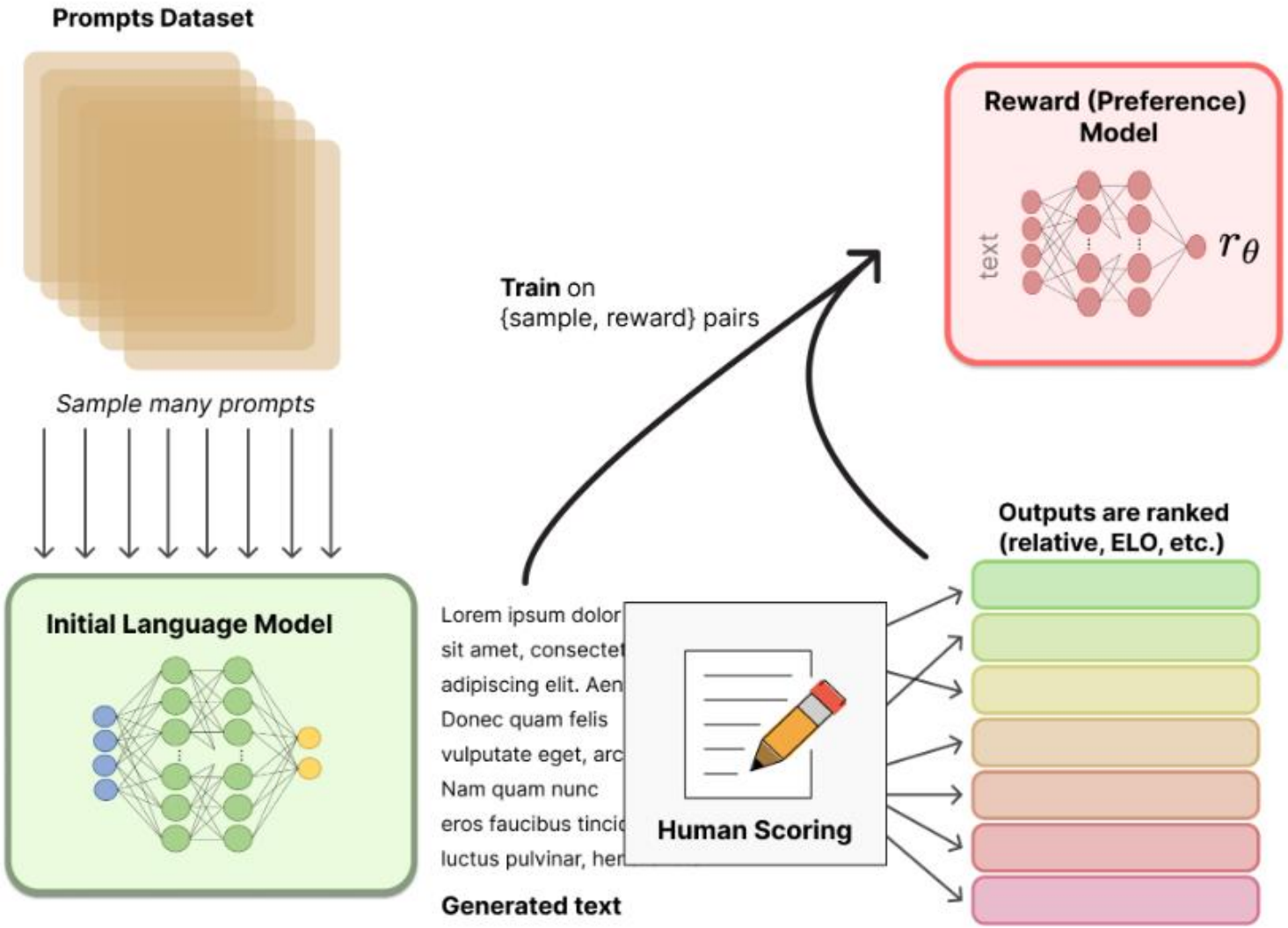
"winning" sample

"losing" sample

s^w should score higher than s^l

- Make sure that the reward model's predictions are accurate enough

Reinforcement Learning from Human Feedback (RLHF): Reward modelling



https://huggingface.co/blog/rlhf

Reinforcement Learning from Human Feedback (RLHF)

Supervised fine-tuning: Imitation

- Fit $\hat{p}(y \approx p^*(y|x)|x)|x)$ for some reference distribution $p^*(y|x)$
- Generative modelling, requires samples from reference policy

RHLF: Optimization

- Find $\hat{p}(y|x)$ such that $\max_p E_p[R(y,x)]$ for a reward $R(y,x)$
- Maximize some reward function we can measure
- LMs are policies, not a mode of some distribution
- The expected reward is non-differentiable

Reinforcement Learning from Human Feedback (RLHF)

Policy gradient (REINFORCE algorithm):

- Rewrite using log-derivative trick, approximate with Monte Carlo samples:

$$\theta_{t+1} = \theta_t + \alpha \frac{1}{m} \sum_{i=1}^{m} R(s_i) \nabla_{\theta_t} \log p_{\theta_t}(s_i)$$

- However, the variances are too high
- RLHF uses the PPO (Proximal Policy Optimization) algorithm, which optimizes the following objective:

$$L(s, a, \theta_k, \theta) = \min \left(\frac{\pi_\theta(a|s)}{\pi_{\theta_k}(a|s)} A^{\pi_{\theta_k}}(s, a), \quad \text{clip} \left(\frac{\pi_\theta(a|s)}{\pi_{\theta_k}(a|s)}, 1 - \epsilon, 1 + \epsilon \right) A^{\pi_{\theta_k}}(s, a) \right)$$

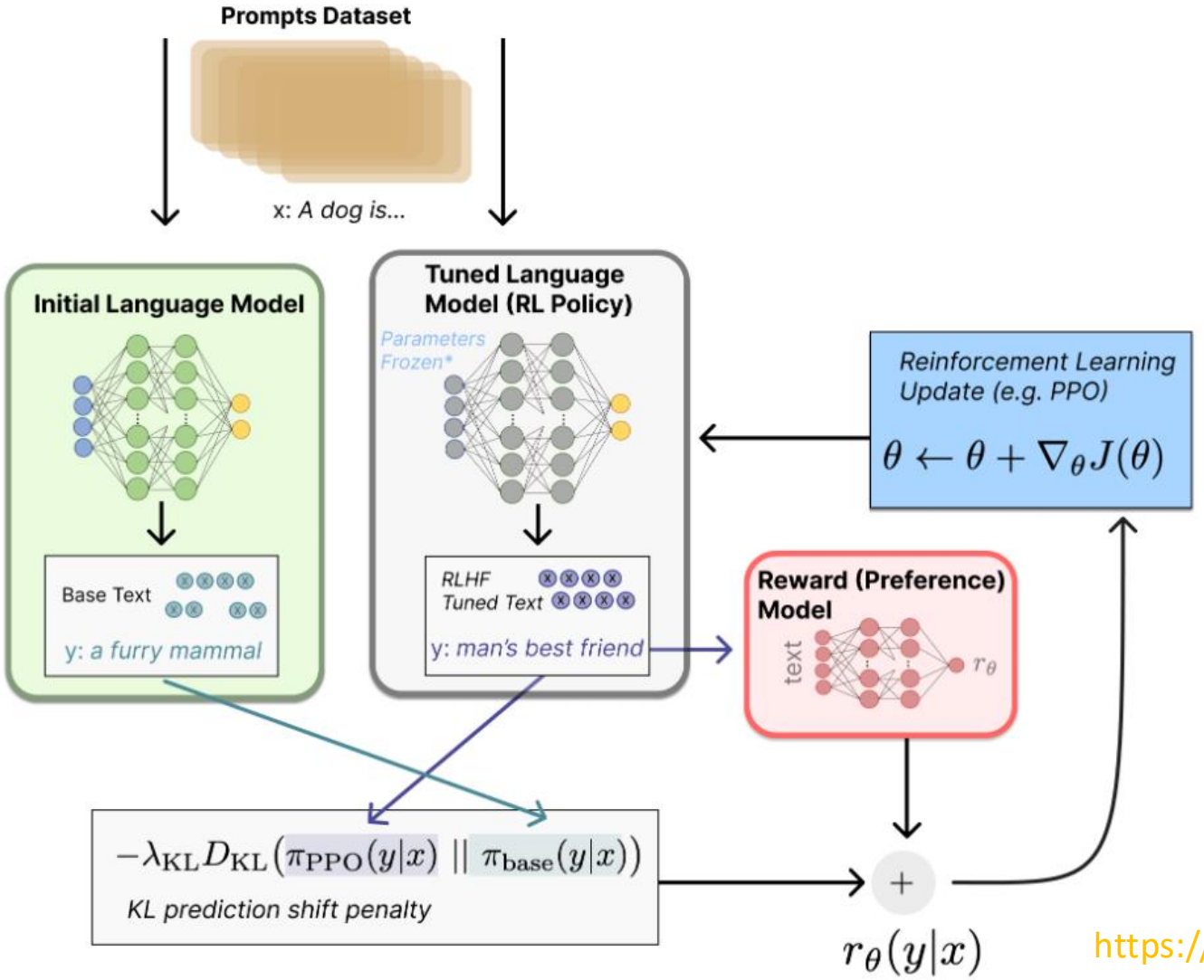
Reinforcement Learning from Human Feedback (RLHF)

PPO

- If the model is purely optimized for the reward, it may generate non-sensical text as output
- Full reward includes a KL divergence term to prevent diverging too far from the pretrained model

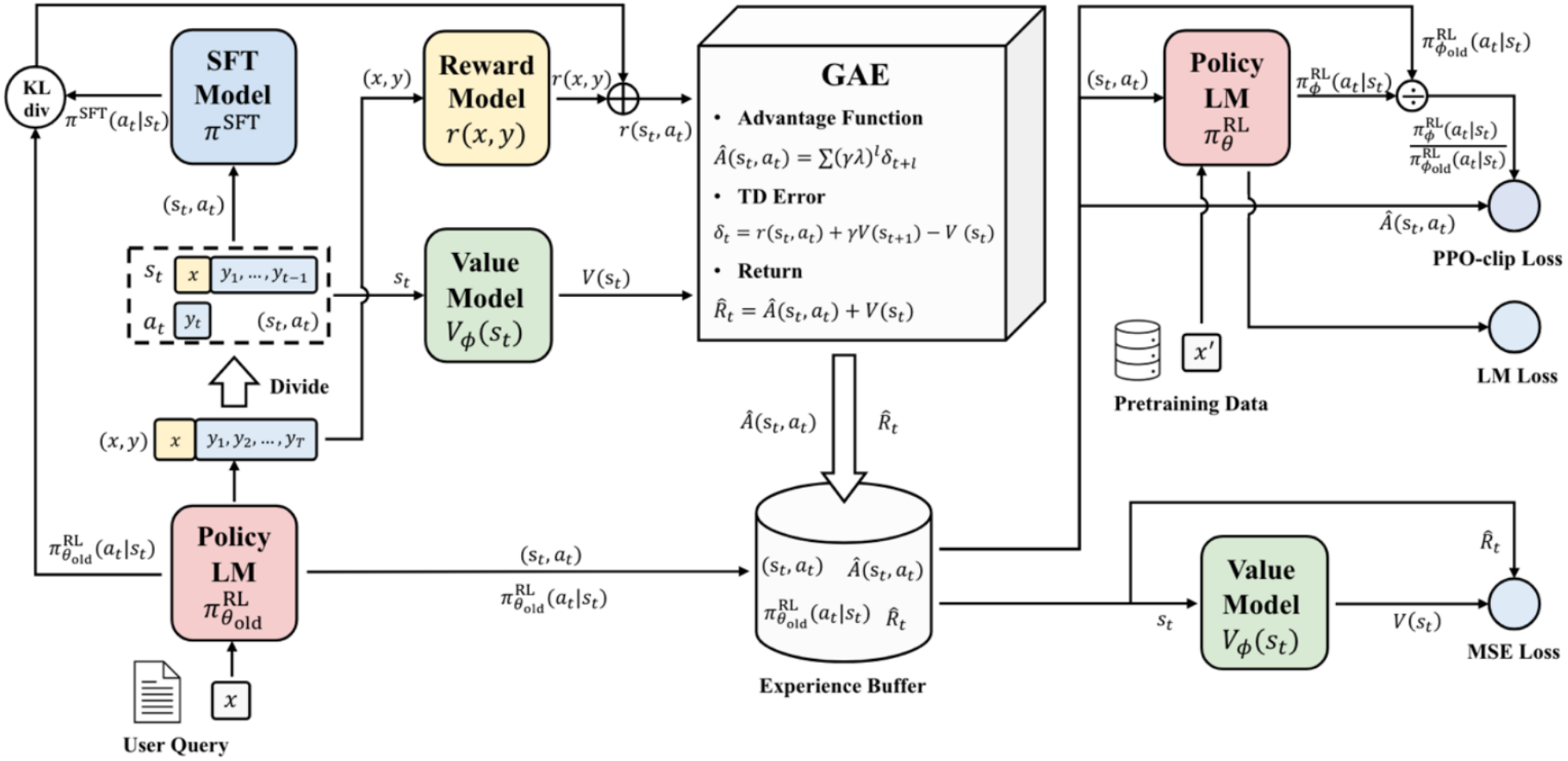
$$R(s) = RM_{\phi}(s) - \underbrace{\beta \log \left(\frac{p_{\theta}^{RL}(s)}{p^{PT}(s)} \right)}$$

Reinforcement Learning from Human Feedback (RLHF): RL fine-tuning



https://huggingface.co/blog/rlhf

Reinforcement Learning from Human Feedback (RLHF)



Reinforcement Learning from Human Feedback (RLHF)

- Stylistic changes after RHLF

Instruction: What are the five most common causes of stress among people?

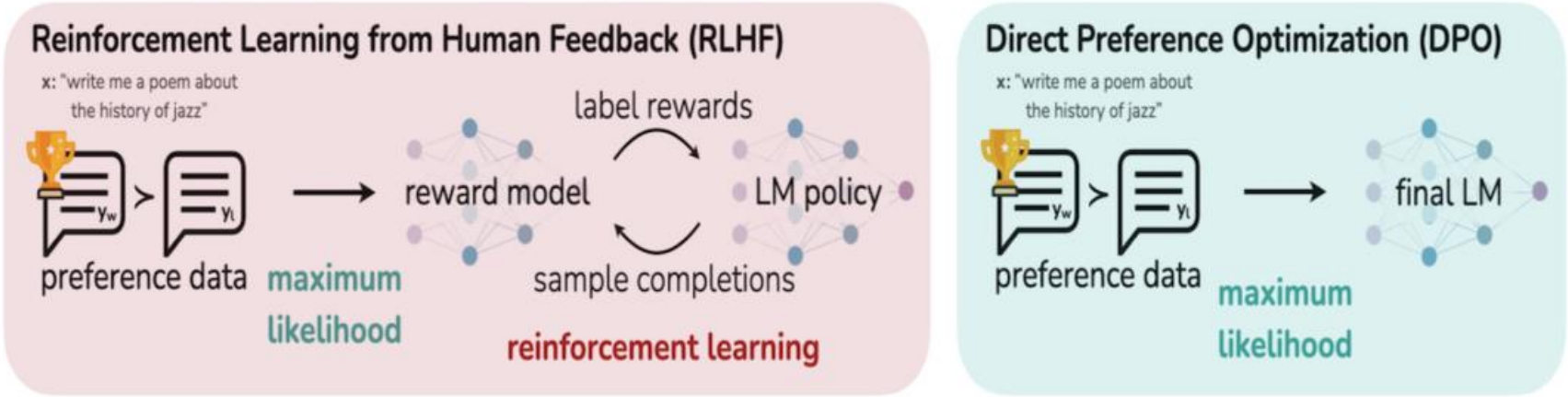
Alpaca (Finetuned w/ 10k data): The five most common causes of stress among people are work, money, relationships, health, and family

Alpaca + PPO: The five most common causes of stress among people are:

1. Pressure at work: Whether it is due to tight deadlines, high expectations, or an overwhelming workload, pressure at work can be a major source of stress
2. Money problems: Financial issues can be a significant source of stress, from not being able to make ends meet to struggling with debt.
3. Unhealthy lifestyle: Poor diet, lack of exercise, and sleep deprivation can all contribute to stress.
4. Unhappiness at home: Problems in relationships, conflicts with family members, and other domestic issues can be a major source of stress.
5. Uncertainty: Excessive worry and anxiety about the future, health issues, or other unknowns can be a major cause of stress

Direct Preference Optimization (DPO)

- Can we learn from preferences in a simpler way?
- Get rid of the reward model and complications from RL
- Instead:
  - Take gradient steps on log-loss of preferred examples
  - Take (weighted) negative gradient steps on undesired examples



Direct Preference Optimization (DPO)

- Goal is to optimize

$$\max_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi_\theta(y|x)}\left[r_\phi(x,y)\right] - \beta\mathbb{D}_{\mathrm{KL}}\left[\pi_\theta(y \mid x) \,||\, \pi_{\mathrm{ref}}(y \mid x)\right]$$

- The "implied reward" is

$$r(x,y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x).$$

- The DPO objective can be derived as:

$$\mathcal{L}_{\mathrm{DPO}}(\pi_\theta; \pi_{\mathrm{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\mathrm{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\mathrm{ref}}(y_l \mid x)}\right)\right]$$

Direct Preference Optimization (DPO)

- The DPO update scales the difference between the gradients of the winning and losing samples by the “prediction error” of the implied reward model

$$\nabla_\theta \mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) =$$

$$-\beta \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\Bigg[\underbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}_{\text{higher weight when reward estimate is wrong}}\Bigg[\underbrace{\nabla_\theta \log \pi(y_w \mid x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_\theta \log \pi(y_l \mid x)}_{\text{decrease likelihood of } y_l}\Bigg]\Bigg],$$

$$\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$$

Reinforcement Learning from Human Feedback (RLHF)

Limitations:

- Chatbots are rewarded to produce responses that seem authoritative and helpful, regardless of truth
  - This can result in making up facts, and hallucinations
- Human preferences are unreliable, models of human preferences even more so
- Optimizing for rewards may lead to overfitting
- It has been shown that over time, RLHF can make the underlying model worse
- Whose values are being modelled?

Reasoning: Explaining Decisions

- Text rationales: Generate a natural language explanation of a model's behaviour or prediction
- Language models often work better when rationales are included
- Rationales or explanations can guide the model to perform reasoning to find a solution
- Example: E-SNLI

Premise: An adult dressed in black holds a stick.
Hypothesis: An adult is walking away, empty-handed.
Label: contradiction
Explanation: Holds a stick implies using hands so it is not empty-handed.

Reasoning: Explaining Decisions

- Example: Solving math word problems

Problem 2:
Question: From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings?
Options: A) 2/1223 B) 1/122 C) 1/221 D) 3/1253 E) 2/153
Rationale: Let s be the sample space.
Then n(s) = 52C2 = 1326
E = event of getting 2 kings out of 4
n(E) = 4C2 = 6
P(E) = 6/1326 = 1/221
Answer is C
Correct Option: C

Chain-of-thought prompting

- Chain-of-thought uses natural language as a scaffold for “reasoning"
- For math: relies on the fact that LLMs can do single steps of arithmetic okay. Builds on that to do multi-step problems
- For QA: many problems involve reasoning decompositions, e.g., “What’s the capital of the country where Aristotle lived?” ->

  country = “country where Aristotle lived”

  return “What’s the capital of [country]”
- For other tasks: capture the kinds of behavior written in rationales

Chain-of-thought prompting

- Few-shot prompting where the in-context examples contain explanations
- Answer is not generated directly, but after an explanation that "talks through" the reasoning

Context: *Christopher agrees with Kevin.* […] **Q**: *Who hangs out with a student?*

Mary**, **because Mary hangs out with Danielle and Danielle is a student.

Predict-explain: answer **is not** conditioned on output explanation (original E-SNLI LSTM)

Context: *Christopher agrees with Kevin.* […] **Q**: *Who hangs out with a student?*

Because Mary hangs out with Danielle and Danielle is a student**, the answer is **Mary.

Explain-predict: answer is conditioned on output explanation (Chain of Thought)

Chain-of-thought prompting

Input:

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

...

Q: John takes care of 10 dogs. Each dog takes .5 hours a day to walk and take care of their business. How many hours a week does he spend taking care of dogs?
A:

Model output:

John takes care of 10 dogs. Each dog takes .5 hours a day to walk and take care of their business. So that is 10 x .5 = 5 hours a day. 5 hours a day x 7 days a week = 35 hours a week. The answer is 35 hours a week. ✔

Wei et al. (2022)

Chain-of-thought prompting

- Zero-shot chain-of-thought: “Let’s think step by step”

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let’s think step by step.***

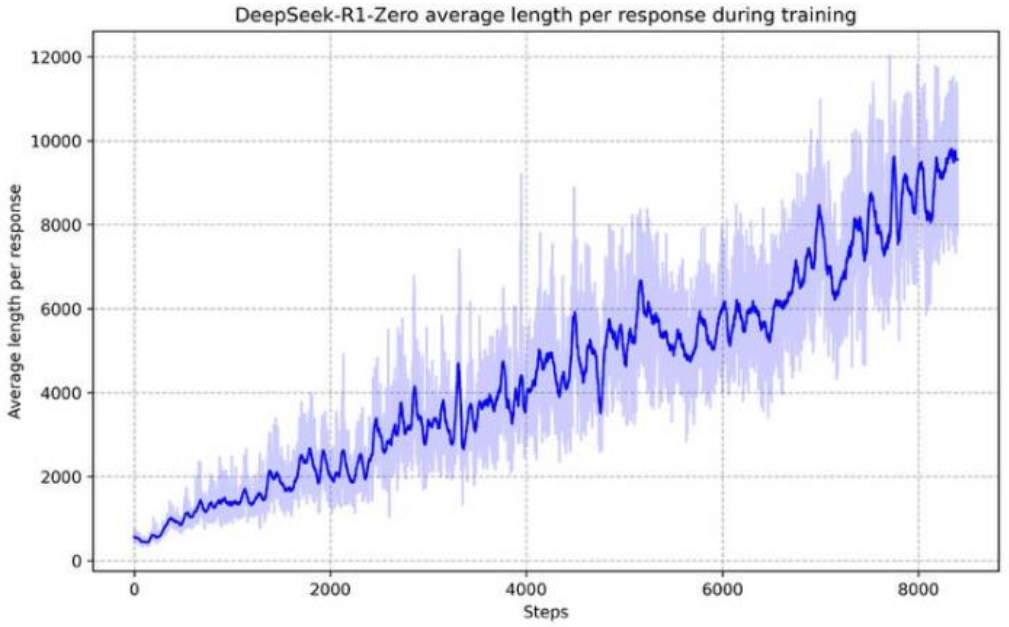
(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Reinforcement Learning from Verifiable Rewards (RLVR)

- RLHF can be used to train an LLM chat model, but not the capabilities of recent models
- The main approach to improve the model further is to learn from tasks where the rewards can be calculated by a verifier (e.g. solving a math problem) rather than human feedback
- Through RL training the model can learn to generate long output sequences, including multiple “reasoning” steps, that will help it to generate the correct final answer

Reinforcement Learning from Verifiable Rewards (RLVR)

- Standard RL at first did not succeed with this
- The **GRPO** algorithm (Group Relative Policy Optimization) was shown to be more suitable for this, in particular for scaling to longer sequences



Test time scaling

- The longer the model “thinks” during inference, the more performance improves

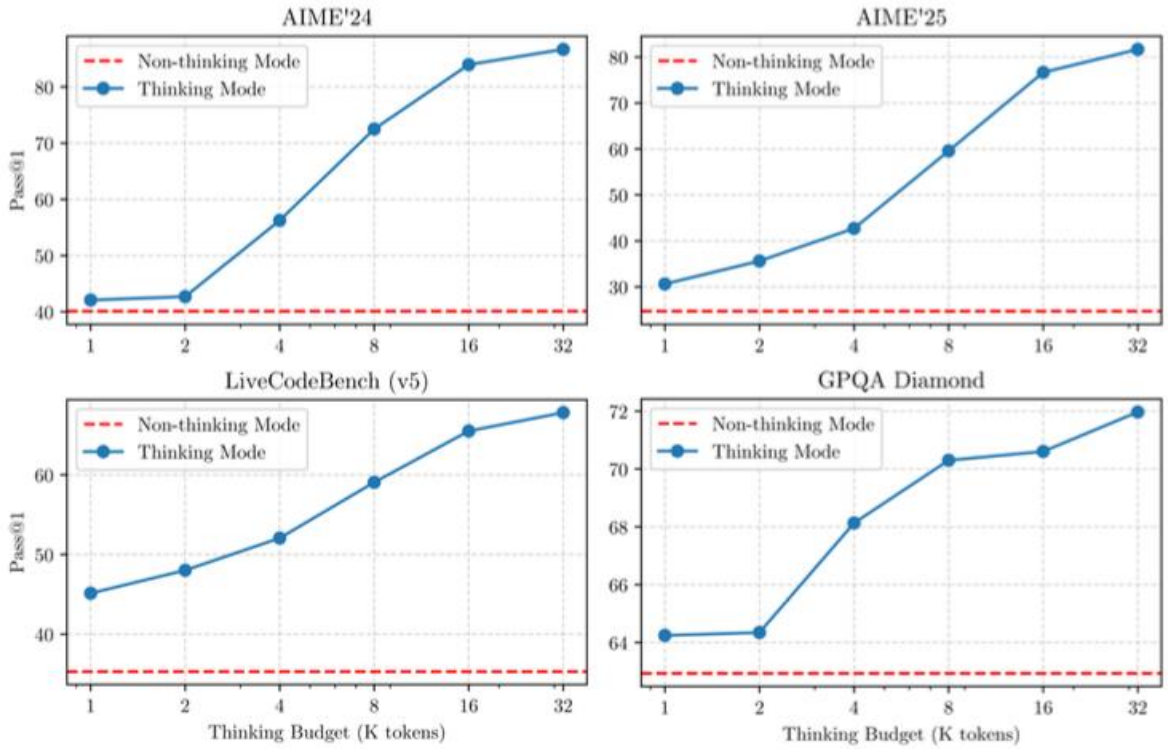


Figure 2: Performance of Qwen3-235B-A22B with respect to the thinking budget.